

JKBOSE 2026

CLASS 11TH

#XUV 

CHEMICAL THERMODYNAMICS

LECTURE-4

CHEMISTRY

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ON
TOP.



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Topics To Covered

- 1 Internal Energy
- 2 Enthalpy
- ★ 3 First Law of Thermodynamics

REVISION

Substance

10 kg
20 kg

10 mol
20 mol



Heat

q.

Path Function

Extensive

Joule

Absorbed = +q
release = -ve.

Heat Capacity

$$C = \frac{q}{\Delta T}$$

$$U_{\text{unit}} = \text{J K}^{-1}$$

Path function

Extensive property

Specific Heat Capacity

$$C_s = \frac{q}{\Delta T \times m}$$

$$U_{\text{unit}} = \text{J K}^{-1} \text{g}^{-1}$$

State Function

Intensive property

Molar heat capacity

$$C_m = \frac{q}{\Delta T \times n}$$

$$U_{\text{unit}} = \text{J K}^{-1} \text{mol}^{-1}$$

1 mole

Internal Energy (U or E)

"It is defined as sum total of all the forms of energies of a system"

It is denoted by U or E

i.e. $U = \text{K.E} + \text{P.E} + \text{Vibrational Energy} + \dots$

Gravitation energy is not included.

It is a state function & extensive property.

In thermodynamics we usually talk about change in energy (Δ).

• Internal energy as a state function $\Rightarrow$

Consider a reaction



Let U_R is the internal energy of reactant & U_P is the internal energy of product.

$$\therefore \Delta U = U_P - U_R$$

$\Delta U = +ve$, when $U_P > U_R$.

& $\Delta U = -ve$, when $U_P < U_R$.

i.e. ΔU depends upon U_P & U_R value.

Hence, Internal energy is a state function.



First Law of Thermodynamics (FLOT)



It was given by Clausius in 1850.

It states that

"Energy can neither be created nor be destroyed only it can be transferred from one form to another"

or

"Total energy of an isolated system ^{always} remains constant"

Example $\rightarrow$

i) Photosynthesis $\rightarrow$ We convert light energy to chemical energy.

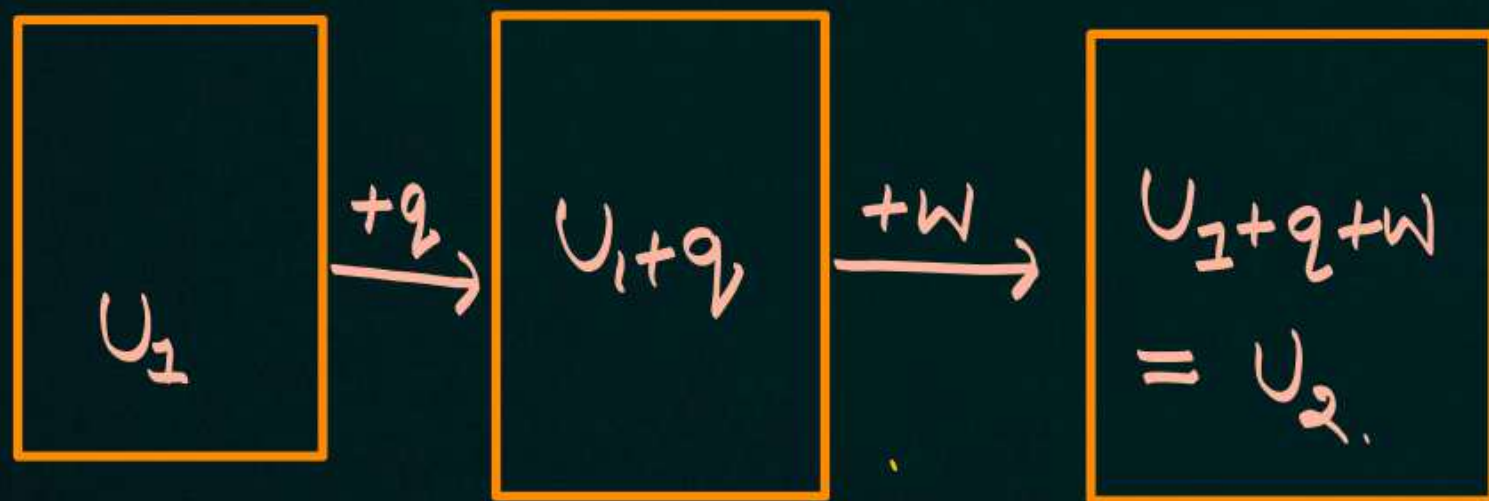
ii) Burning of Coal $\rightarrow$ We convert chemical energy into heat & light energy.

iii) Electrical bulb $\rightarrow$ We convert electrical energy into light energy.

First Law of Thermodynamics (FLOT)



DERIVATION Suppose a system is present with internal energy U_1 , internal energy can be changed only by providing heat to the system or by doing work on the system.



Therefore, final internal energy i.e $U_2 = U_1 + q + W$.

$$\therefore U_2 - U_1 = q + W.$$

$$\Delta U = q + W$$

$$\text{or } \Delta U = q + (-P\Delta V).$$

$$\text{or } \Delta U = q - P\Delta V.$$

$$\text{or } q = \Delta U + P\Delta V.$$

This is the mathematical equation of first law of thermodynamics.

Enthalpy

It is defined as part of internal energy of a system that can be converted into heat is called enthalpy".

It is denoted by 'H'

Mathematically $\Rightarrow$

$$H = U + PV$$

i.e. From equal ①

$$\Delta H = \Delta U + P\Delta V$$

Heat absorbed at constant P.

According to first law of thermodynamics

$$q_p = \Delta U + P\Delta V \quad \text{--- ①}$$

$$q_p = U_2 - U_1 + P(V_2 - V_1)$$

$$q_p = U_2 - U_1 + PV_2 - PV_1$$

$$q_p = U_2 + PV_2 - U_1 - PV_1$$

$$q_p = (U_2 + PV_2) - (U_1 + PV_1)$$

$$q_p = H_2 - H_1$$

$$q_p = \Delta H$$

Heat absorbed at constant pressure is equal to the change in enthalpy.

Enthalpy



Heat absorbed at constant Volume.

According to first law of thermodynamics. (q_v)

$$q_v = \Delta U + P\Delta V \quad \text{but Volume is constant} \\ \text{i.e. } \Delta V = 0$$

$$q_v = \Delta U$$

Heat absorbed at constant volume is equal to the change in internal energy.

★

$$q_p = \Delta H$$

$$q_v = \Delta U$$

Relationship between q_p & q_v

As we know,

$$\Delta H = \Delta U + P\Delta V$$

$$\Delta H = \Delta U + P(V_2 - V_1)$$

$$\Delta H = \Delta U + PV_2 - PV_1$$

According to ideal gas equation ($PV = nRT$)

$$PV_2 = n_2RT \text{ \& } PV_1 = n_1RT$$

Therefore

$$\Delta H = \Delta U + n_2RT - n_1RT$$

$$\text{or } \Delta H = \Delta U + (n_2 - n_1)RT$$

$$\text{or } \Delta H = \Delta U + \Delta n_g RT$$

Δn_g is the change in no. of moles of a gas in a chemical equation.

\& we also know that

$$q_p = \Delta H \text{ \& } q_v = \Delta U$$

Therefore

$$q_p = q_v + \Delta n_g RT$$

Revision.

$$\begin{aligned}\Delta H &= \Delta U + P\Delta V \\ &= \Delta U + P(V_2 - V_1) \\ &= \Delta U + PV_2 - PV_1\end{aligned}$$

Ideal gas equation

$$PV_2 = n_2 RT \text{ \& } PV_1 = n_1 RT.$$

$$\begin{aligned}\Delta H &= \Delta U + n_2 RT - n_1 RT \\ &= \Delta U + (n_2 - n_1) RT.\end{aligned}$$

$$\Delta H = \Delta U + \Delta n_g RT.$$

or $q_p = q_v + \Delta n_g RT.$



$$\begin{aligned}\Delta n_g &= n_2 - n_1 \\ &= n_p - n_r \\ &= 3 - 2 \\ &= \textcircled{1}.\end{aligned}$$

QUESTION



State first law of Thermodynamics. Derive its mathematical relation.

(PYQ)

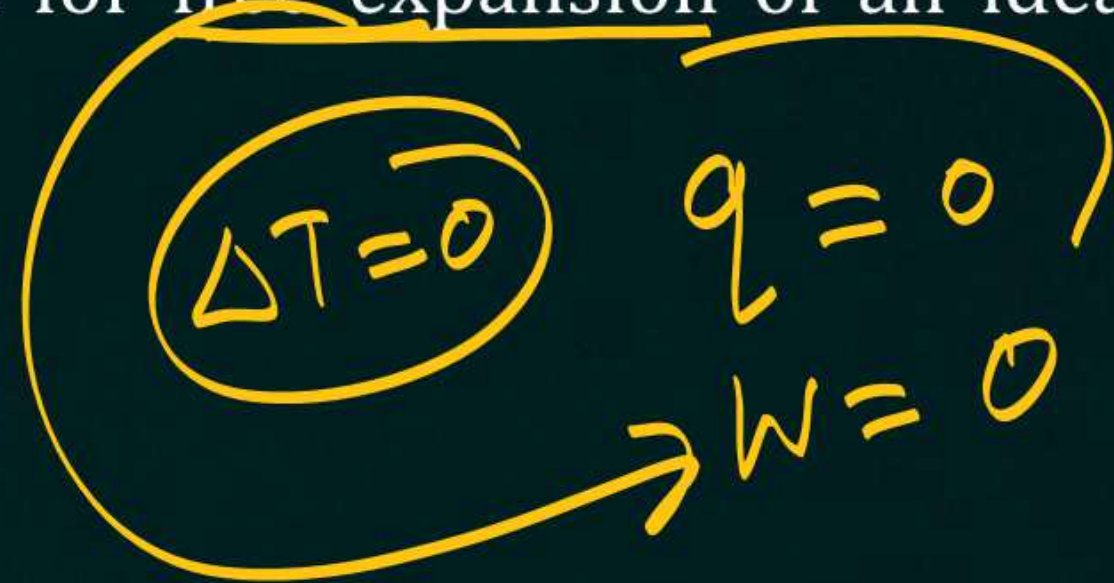
QUESTION



State first of thermodynamics and give a mathematical expression for it. Mention two examples to support the law. (PYQ).

QUESTION

In an adiabatic process, no transfer of heat takes place between system and surroundings. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following.



$$\Delta T = 0 \quad q = 0 \quad \rightarrow \quad W = 0$$

1 $q = 0, \Delta T \neq 0, W = 0$

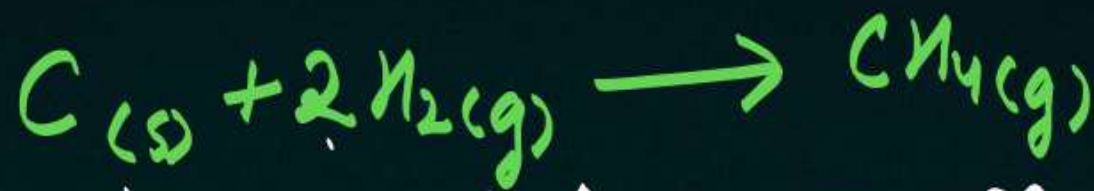
2 $q \neq 0, \Delta T = 0, W = 0$

3 $q = 0, \Delta T = 0, W = 0$

4 $q = 0, \Delta T < 0, W \neq 0$

QUESTION

Given



$$\Delta n_g = n_p - n_r = 1 - 2 = -1$$

$\Delta_f U^\ominus$ of formation of $CH_4(g)$ at certain temperature is -393 kJ mol^{-1} . The value of $\Delta_f H^\ominus$ is.

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\begin{aligned}\Delta H &= -393 - 1 RT \\ &= -393 - RT\end{aligned}$$

$$\Delta U = -393 \text{ kJ mol}^{-1}$$

$$\Delta H = ?$$

$$\Delta H = \Delta U + P\Delta V$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = 298 \text{ K}$$

1 Zero

2 $< \Delta_f U^\ominus$

3 $> \Delta_f U^\ominus$

4 Equal to $\Delta_f U^\ominus$

QUESTION



Work done by the system = -ve

(U)
What is the change in the energy of system if 500 cal of heat energy are added to a system and system does 350 cal of work on the surroundings?

$$\begin{aligned}\Delta U &= q + W \\ &= 500 + 350 \\ &= \underline{850 \text{ cal}} \\ &= 500 + (-350) \\ &= 500 - 350 \\ &= \underline{150}\end{aligned}$$

$$\begin{aligned}q &= 500 \text{ cal} \\ W &= -350 \text{ cal} \\ \Delta U &= ?\end{aligned}$$

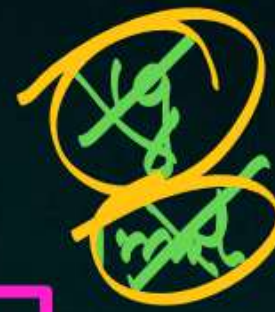
1 -150 cal

2 +150 cal

3 + 850 cal

4 -850 cal

QUESTION



The heat required to raise the temperature of body by 1 K is called.



1 Specific heat

2 Heat or Thermal capacity

3 Water equivalent

4 None of these.

REVISTE KAR LENA.

Thank
You